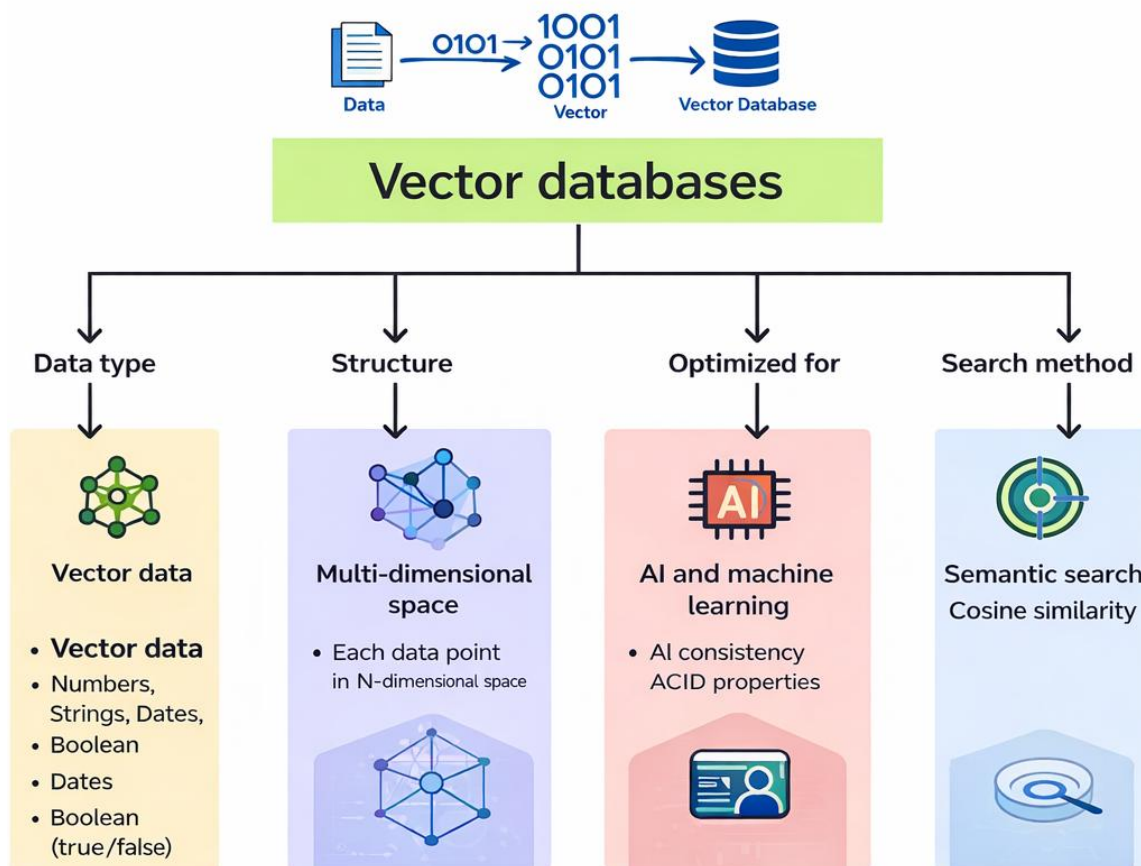
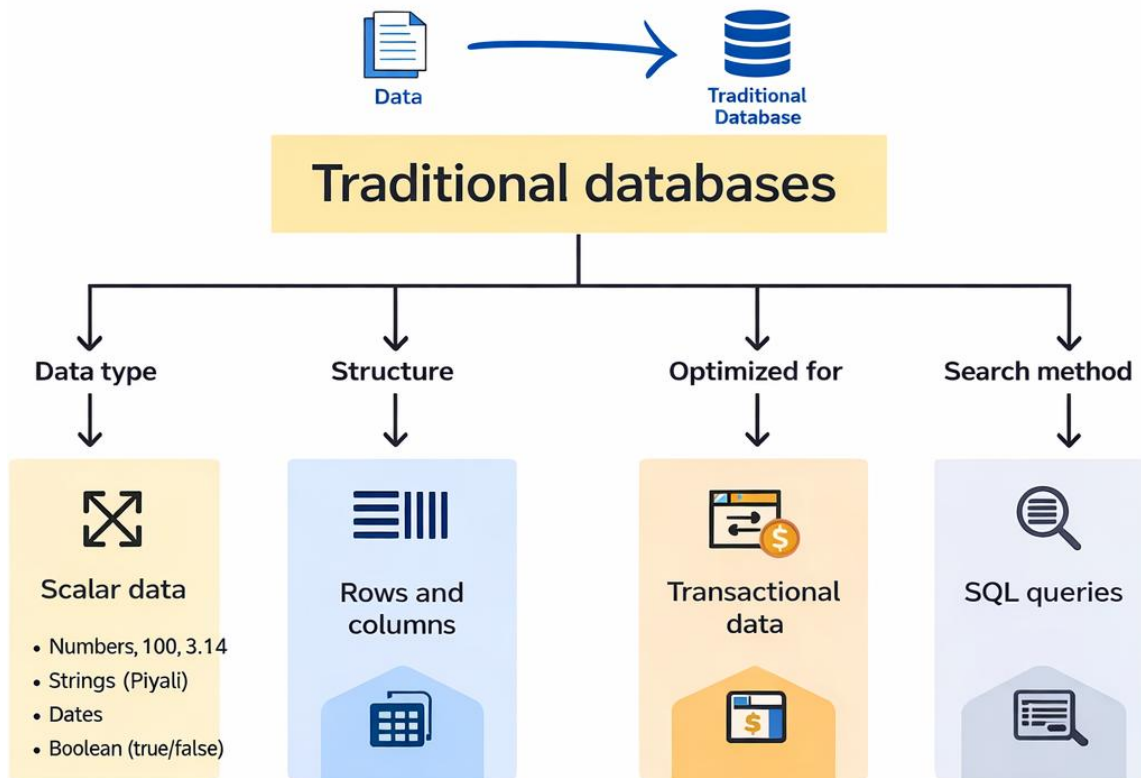


Traditional Databases vs Vector Databases



◆ Traditional Databases

Examples:

- MySQL
- PostgreSQL
- Oracle Database

1 Data Type → Scalar Data

- Numbers (100, 3.14)
- Strings ("Piyali")
- Dates
- Boolean (true/false)

Structured, exact values.

2 Structure → Rows & Columns

- Data stored in tables
- Fixed schema
- Example:

id	name	age
1	Piyali	25

3 Optimized For → Transactional Data (OLTP)

- Banking systems
- E-commerce orders
- Inventory management
- CRUD operations (Create, Read, Update, Delete)

High consistency, ACID properties.

Search Method → SQL Queries

Example:

```
SELECT * FROM users WHERE age = 25;
```

 Exact match search

- =
- LIKE
- JOIN
- WHERE

They are NOT designed for semantic similarity.

◆ Vector Databases

Examples:

- Qdrant
 - Pinecone
 - Weaviate
-

Data Type → Vector Data (Embeddings)

Instead of storing:

"FastAPI is a web framework"

It stores something like:

[0.213, -0.982, 0.551, 0.112, ...]

These are high-dimensional numeric representations of meaning.

Structure → Multi-Dimensional Space

- Each data point = a point in N-dimensional space
- Similar meanings → closer distance
- Dissimilar meanings → farther apart

Like a 768-dimension coordinate system.

3 Optimized For → AI & Machine Learning

- RAG systems
- Semantic search
- Recommendation systems
- Chatbots
- Image similarity

Perfect for LLM-based apps (like what you’re building with FastAPI + LangChain + Qdrant).

4 Search Method → Similarity Search

Instead of:

WHERE name = 'Piyali'

You do:

- Cosine similarity
- Euclidean distance
- Dot product

Example query meaning:

“Find documents similar to: FastAPI tutorial”

It returns semantically close results — not exact matches.

🔥 Core Difference Summary

Feature	Traditional DB	Vector DB
Data	Scalar	Embeddings
Structure	Tables	Multi-dimensional space
Query Type	Exact match	Similarity search

Feature	Traditional DB	Vector DB
Optimized For	Transactions	AI/ML
Example Use	Banking system	RAG chatbot

💡 Why This Matters for You

Since you're working on:

FastAPI → LangChain → Qdrant → Ollama

You need:

- Traditional DB → store user logs, metadata, configs
- Vector DB → store embeddings for semantic retrieval

👉 They don't compete.

👉 They solve different problems.

👉 Modern AI systems use BOTH.